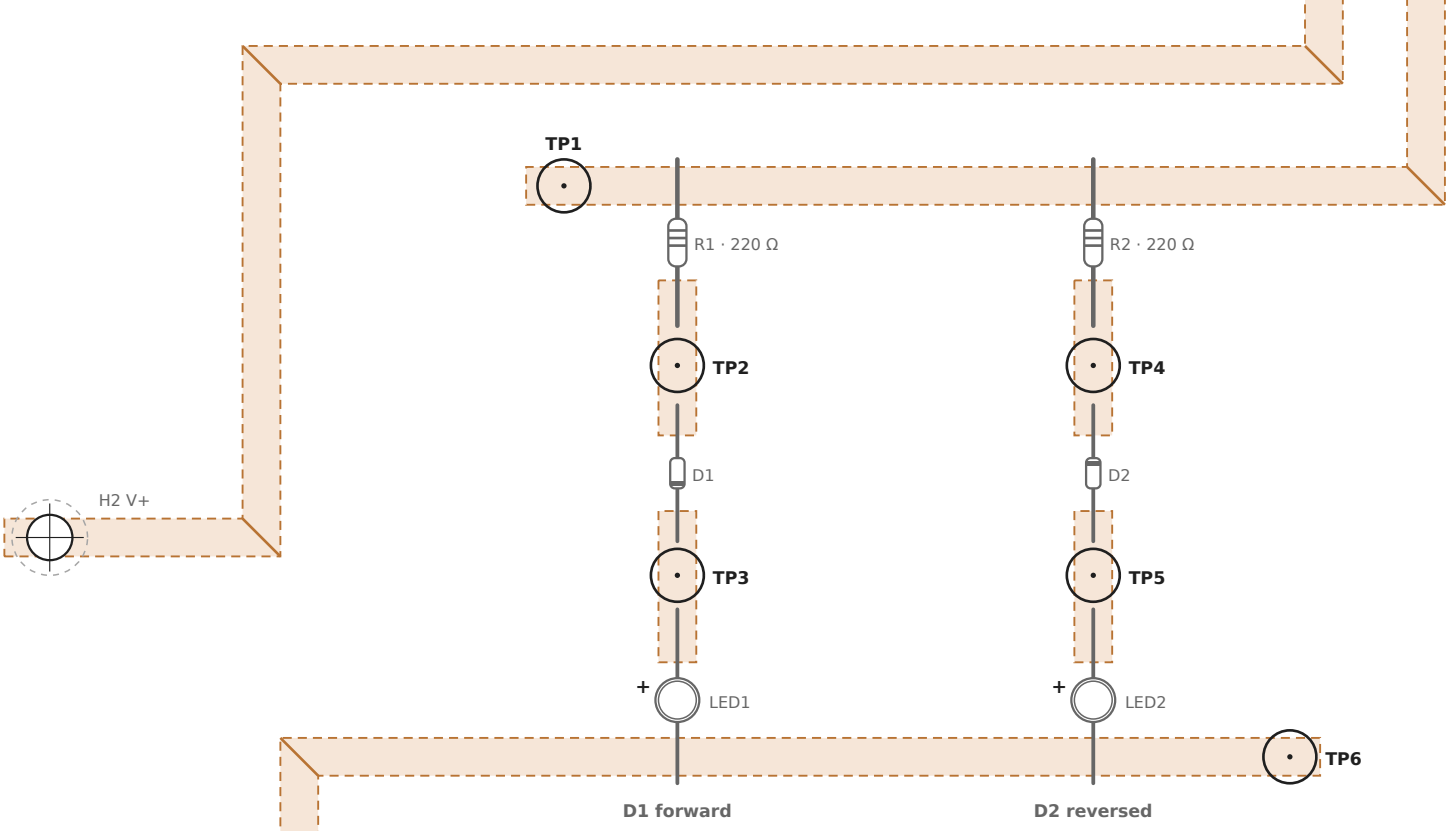
L06 Diodes are one-way valves

A diode lets current through one way only. Which LED gets any?

Draft: tagged steps may change

H1 BUS A · NC

S1
fold the corner down



H2 V+

H3 GND

You need

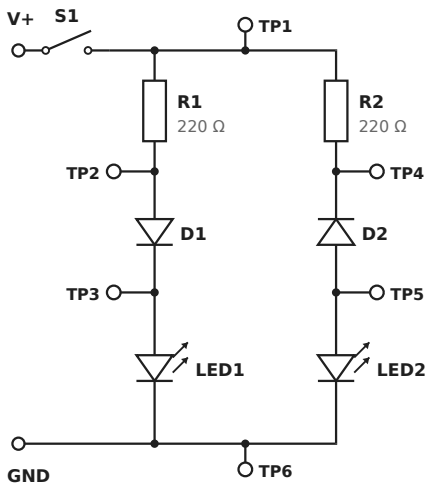
- 2 red LEDs, 5 mm
- 2 resistors, 220 Ω (red-red-brown) **[P]**
- 2 small diodes, 1N4148 (glass, black band)
- 5 mm copper tape
- Power page P1 in the binder; a multimeter

Build it

1. Lay the seven lanes, one piece of tape each. Fold at the creases; never cut.
2. Stick a 10 mm piece of copper tape on the corner flap, S1. **[T0]**
3. Punch the holes and set eyelets in H2 and H3. **[T0]**
4. Place R1 and R2 across their gaps, either way round.
5. Place D1 and D2 **exactly as printed**, black band on the band mark. D2 points the other way on purpose.
6. Place LED1 and LED2 as printed: long leg on the +.
7. Hold each lead flat on its lane with a small patch of copper tape. **[T0]**
8. Bind this page with P1, then fold the corner down to switch on.

[T0] and **[P]** mark choices that may change: **[T0]** after the contact tests, **[P]** once the power supply is settled.

Schematic



The triangle points the way current can flow.

Predict D2 is printed the other way round.

When you fold the corner down, which LEDs will light? Circle one for each.

LED1 lights / stays dark **LED2** lights / stays dark

Why do you think so?

Measure Hold the corner down. Black probe on TP6.

Diode	Before	After	Across
D1 TP2, TP3	____ V	____ V	____ V
D2 TP4, TP5	____ V	____ V	____ V

Across = before − after

Explain D1 takes about 0.7 V. Where does the rest of the supply go?

Why is LED2 dark, and why is there so much voltage across D2?

H4 BUS B · NC

Print at 100 %: this bar should measure 100 mm.